

A glimpse of stereometry

Bojan Bašić, bbasic@gmail.com

1. Let $\triangle ABC$ be an equilateral triangle. Points M and N are chosen on lines BC and AC , respectively. Prove that it is possible to form a triangle of the segments AM , BN , and MN .
2. Let $\triangle ABC$ and a point P be given in the plane, where P is distinct from A , B , C . Let X , Y , Z be the reflections of P with respect to the midpoints of BC , CA , AB , respectively. Show that the lines AX , BY , CZ pass through a common point.
3. Let an equilateral $\triangle ABC$ and a line l not parallel to any of its sides be given. The lines AB , AC , and BC intersect the line l at the points R , Q , and P , respectively. Let k_1 be the circle whose center is P and that passes through A , let k_2 be the circle whose center is Q and that passes through B , and let k_3 be the circle whose center is R and that passes through C . Prove that the circles k_1 , k_2 , and k_3 have a common point.
4. Let an equilateral triangle $\triangle ABC$ be given and let S be its center. A line l contains S and intersects AC and BC at K and L , respectively. Let P be a point such that $AL = PL$ and $BK = PK$. Prove that the distance from the point P to the line l is constant, regardless of the positions of l and P .
5. Three circles of different radii, none of which is completely contained in another, are given in the plane. For each pair of circles, mark the intersection point of their common external tangents. Prove that the three marked points are collinear.
6. Let three circles be given in the plane. Each pair of them intersects in two points, but they do not all pass through the same point. Prove that the three common chords of the three pairs of circles pass through a common point.
7. (Desargues's theorem) Let $\triangle ABC$ and $\triangle A_1B_1C_1$ be given in the plane. Let X be the intersection point of the lines BC and B_1C_1 , let Y be the intersection point of the lines AC and A_1C_1 , and let Z be the intersection point of the lines AB and A_1B_1 . Prove that the points X , Y , and Z are collinear if and only if the lines AA_1 , BB_1 , and CC_1 pass through a common point.
8. (Menelaus's theorem) Let a $\triangle ABC$ be given, and let X , Y , Z be points on the lines AB , BC , CA , respectively. Suppose that X , Y , Z are distinct from A , B , C , and that either only one of them or all three of them are on the sides of the $\triangle ABC$ (that is, not on side extensions). Then X , Y , Z are collinear if and only if

$$\frac{AX}{XB} \cdot \frac{BY}{YC} \cdot \frac{CZ}{ZA} = 1.$$

9. Let $\triangle ABC$ be an equilateral triangle, and P a point on its incircle. Prove that $AP^2 + BP^2 + CP^2$ is constant.
10. Given in the plane are an equilateral triangle of side a , and a point whose distances from its vertices are b , c , and d . Prove:

$$3(a^4 + b^4 + c^4 + d^4) = (a^2 + b^2 + c^2 + d^2)^2.$$
11. Four cars drive in the Sahara desert (an infinite plane) at arbitrary but constant velocities and directions. No two of their directions are parallel and no three are passing through the same point. It is known that the car A at some point in time meets the car B (though let's pretend they drive through each other without crashing). The exact same thing is also known for the pairs $\{A, C\}$, $\{A, D\}$, $\{B, C\}$, and $\{B, D\}$.
 - a) Is it also true for the pair $\{C, D\}$?
 - b) Prove that, at any point in time, all the cars are collinear.
12. Is it possible to cover a circular hole of diameter 10 by 9 long enough planks of width 1?
13. A regular hexagon is tiled with diamonds (pairs of unit triangles glued together along an edge). Diamonds come in three varieties, depending on orientation. Prove that precisely the same number of each variety must appear in the tiling.
14. Two thieves stole an open necklace with beads made from 3 different types of gems. They want to divide the necklace in a fair way between them, that is, in such a way that each thief gets half of the stones of each type of gem. They further want to do so cutting the necklace as few times as possible. How many cuts at minimal do they need?
15. Prove that any closed curve in the plane contains four points that are vertices of a rectangle.
16. The projections of a solid body onto two planes are perfect disks. Prove that they have the same radius.
17. Let P be a polyhedron with red and green faces such that every red face is surrounded by green ones, but the total red area exceeds the total green area. Prove that you cannot inscribe a sphere in P .
18. To each face of a polyhedron we associate an out-pointing vector perpendicular to that face, with length equal to the area of that face. Prove that the sum of these vectors is zero.
19. Let $ABCD$ be a regular tetrahedron. Points M and N are given in the planes BCD and ACD , respectively. Prove that it is possible to form a triangle of the segments AM , BN , and MN .
20. Four spheres of different radii, none of which is completely contained in another, are given in space. For each pair of spheres, mark the intersection point of their common external tangents. Prove that the six marked points are coplanar.